GTS123 Introduction to Computers and Programming

Lab 10: Problem 2

Write a Python program that solves the following system of linear equations using inverse matrix.

$$C_1x + C_2y + C_3z = C_4$$
$$C_5x + C_6y + C_7z = C_8$$
$$C_9x + C_{10}y + C_{11}z = C_{12}$$

The program accept the coefficient values from a user. The program prints **Unable to find a solution** when it cannot find an inverse of the coefficient matrix. The output should look similar to the following.

```
Input C1: 5
Input C2: -2
Input C3: 3
Input C4: 20
Input C5: 2
Input C6: -4
Input C7: -3
Input C8: -9
Input C9: 1
Input C10: 6
Input C11: -8
Input C12: 21
Solution:
x = 5.00
y = 4.00
z = 1.00
```

```
Input C1: 1
Input C2: 2
Input C3: 3
Input C4: 6
Input C5: 4
Input C6: 5
Input C7: 6
Input C8: 15
Input C9: 7
Input C10: 8
Input C11: 9
Input C12: 24
Unable to find a solution
```

Note: To pass the test, your program must produce the exact output as shown above, including spaces and line breaks.